MAYER-VIETORIS DISTINGUISHED TRIANGLE FOR EXTENSION BY ZERO ALONG THE INCLUSIONS OF TWO OPEN SUBSPACES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof ends with the author's own '(computation omitted)' after naming the stalk computation; the parenthetical is reproduced verbatim. The omitted computation is a bounded stalk check which the named reference carries out, and the identification of the triangle with a short exact sequence is complete as written.

Editorial note. The counted spans carry 4 navigation stubs and no bibliography citation.

Editorial note. The preamble is the chapter's retained minimal preamble (sources/stacks-cohomology/retained-preamble.tex lines 1-81), a verbatim excerpt of the native preamble.

Editorial note. The label lemma-exact-sequence-lower-shriek is defined only in cohomology.tex, so no cross-chapter restatement had to be weighed for this card.

Allowed standard prerequisites (editorial).

- Restriction and extension by zero along an open immersion are computed without deriving. Same chapter, Section section-properties-K-injective, invoked at proof lines 8098-8101. Supplies that an arbitrary complex represents the derived functors involved.
- Stalk description of extension by zero for modules. Sheaves, Lemma sheaves-lemma-j-shriek-modules, invoked at proof lines 8113-8114. Supplies the termwise exactness check of the displayed sequence.
- Distinguished triangle attached to a short exact sequence of complexes. Derived Categories, Section derived-section-canonical-delta-functor and Lemma derived-lemma-derived-canonical-delta-functor, invoked at proof lines 8102-8106. Supplies the triangle once the short exact sequence is in hand.

Source reference. Original lemma 0152: sources/stacks-chow/tags.txt, source lines 1493–1493. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original section 014Z: sources/stacks-chow/tags.txt, source lines 1490–1490. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original section 0D5U: sources/stacks-cohomology/pinned-original-source.tex, source lines 7814–7814. Pointer outside the counted span, invoked once

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for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 00A7: sources/stacks-chow/tags.txt, source lines 379–379. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let $(X, \mathcal{O}_X)$ be a ringed space. Let $X = U \cup V$ be the union of two open subspaces. For any object E of $D(\mathcal{O}_X)$ we have a distinguished triangle*

$$j_{U\cap V!}E|_{U\cap V} \to j_{U!}E|_U \oplus j_{V!}E|_V \to E \to j_{U\cap V!}E|_{U\cap V}[1]$$

in $D(\mathcal{O}_X)$.

Proof. We have seen in Section 0D5U that the restriction functors and the extension by zero functors are computed by just applying the functors to any complex. Let $\mathcal{E}^\bullet$ be a complex of $\mathcal{O}_X$-modules representing E. The distinguished triangle of the lemma is the distinguished triangle associated (by Derived Categories, Section 014Z and especially Lemma 0152) to the short exact sequence of complexes of $\mathcal{O}_X$-modules

$$0 \to j_{U\cap V!}\mathcal{E}^\bullet|_{U\cap V} \to j_{U!}\mathcal{E}^\bullet|_U \oplus j_{V!}\mathcal{E}^\bullet|_V \to \mathcal{E}^\bullet \to 0$$

To see this sequence is exact one checks on stalks using Sheaves, Lemma 00A7 (computation omitted). □